

Problem

Are the mug and donut homeomorphic?

Structural Understanding

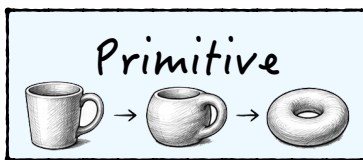
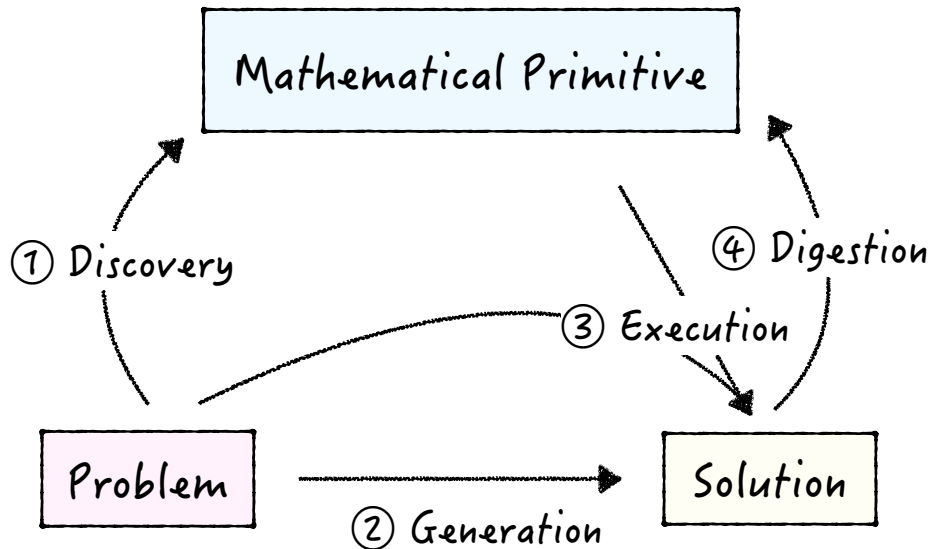
Both have one hole (genus 1).



Solution

1. Construct a map $f: M \rightarrow D$.
2. Verify that f is bijective.
3. Verify continuity of f and f^{-1} .

Four Dimensions of Mathematical Understanding



Primitive

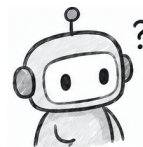


Teacher

Primitive-Guided Reasoning

Internalize

Problem



Student

Student-Generated Reasoning